

YASH RUPANI

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EDUCATION

Oregon State University

M.E. in Computer Science, GPA: 3.87/4

Corvallis, OR

Sep 2023 - Dec 2025

Pandit Deendayal Energy University, India

B.Tech. in Electrical Engineering, GPA: 7.6/10

Gujarat, India

Aug 2017 - Jun 2021

WORK EXPERIENCE

Software Engineering Intern | Python, FastAPI, OCR, Neo4j, Render, Cloudflare

May 2026 - Present

QuickGrants

Remote, USA

- Owned the end-to-end OCR-based document-parsing pipeline, reducing manual data-extraction effort by 80% across 100+ page policy documents at 95% extraction accuracy.
- Optimized technical demo prep time by 15 weekly hours, modeling parsed policy data in a graph database and deploying a cloud-hosted service that turned unstructured text into structured, queryable reports.

Research Assistant | Python, Agentic AI, NLP, Conversational AI Systems

Oct 2025 - May 2026

Oregon State University

Corvallis, OR

- Co-developed a computational framework to operationalize alliance variance, a novel metric modeling emotional attunement and goal alignment across conversational turns in AI-mediated interactions, contributing to a peer-reviewed paper accepted at the [AMCIS 2026 conference](#).
- Prototyped an early-stage agentic AI system that processes de-identified psychiatrist-patient session transcripts to monitor relational-dynamics signals in real time, laying groundwork for systematic evaluation of AI-mediated mental health support systems.

AI Agent RAG/ML Scientist Intern | Python, FAISS, RAG, Flask, Firestore, AWS/GCP

Jul 2025 - Sep 2025

GrantAide

Remote, USA

- Architected a RAG-based grant discovery system, applying system design principles to improve search relevance and retrieval accuracy by 40% over keyword-based search for nonprofits.
- Engineered and deployed multi-cloud backend APIs, reducing response latency by 30% and cross-region latency by 25% while maintaining 99.9% uptime.

Senior Systems Engineer | Postman, REST APIs, CI/CD, AWS EC2, MySQL, Agile

Jun 2021 - Jun 2023

Infosys (Client: Charter Communication)

Maharashtra, India

- Enhanced IVR/IVA service reliability by validating and troubleshooting REST API integrations, analyzing 70+ scenarios per release cycle to identify service, routing, and data-flow issues.
- Reduced manual backend validation effort by about 10 hours per week by automating API and regression workflows and integrating checks into CI/CD pipelines for faster release feedback.
- Supported reliable application deployments by managing and troubleshooting cloud environments, validating behavior across infrastructure and deployment cycles.
- Led root-cause debugging efforts with a 4-5 member engineering team, using data analysis to trace backend integration issues across services and deliver production-ready fixes.

PROJECTS

Shop Pulse: Real-Time E-Commerce Lakehouse | [Live Demo](#) | Python, Spark Streaming, Kafka, Docker, Parquet

- Designed a real-time Lakehouse pipeline grounded in scalable system design practices to power live e-commerce data visualization with sub-second end-to-end latency.
- Implemented a fault-tolerant storage layer, processing hundreds of thousands of simulated transaction records and improving query performance by 20-30% over an unpartitioned baseline.

Crypto Sentinel: Real-Time Market Monitor | [Source Code](#) | Python, FastAPI, Kafka, WebSocket, GitHub Actions

- Devised a microservices-based streaming ingestion service to process live market data, cutting data latency by 40% compared to traditional polling methods.
- Constructed a WebSocket alert system with an auto-triggered CI/CD deployment workflow, processing 500+ events per second to deliver real-time notifications when asset prices crossed user-defined thresholds.

TECHNICAL SKILLS

Languages:

Python, SQL, TypeScript, C, C++, Java

Backend Frameworks & APIs:

Django, Flask, Node.js, RESTful API design, FastAPI, WebSocket, Git

Cloud:

AWS (EC2, S3, Lambda, Athena, DynamoDB), GCP, Docker

Databases:

MySQL, PostgreSQL, Neo4j (Graph DB), Firestore

AI / ML:

RAG Architectures, Vector Databases (FAISS), AI Agents

Tools:

Pega, JIRA, Render, Cloudflare, GitHub Actions, Kubernetes, Apache Spark, Kafka

PUBLICATIONS & CERTIFICATIONS

Publication: Co-author, [Computational Alliance Modeling for Conversational AI](#) (Accepted at AMCIS 2026)

Certifications: [Microsoft Certified: Fabric Data Engineer Associate](#) | [Snowflake Data Warehousing](#) | [Databricks AI Agent](#)